

Federica Stolf

CONTACT INFORMATION	Department of Statistical Science Duke University 214 Old Chemistry, Durham, NC 27708	✉: federica.stolf@duke.edu web: https://federicastolf.netlify.app
RESEARCH INTERESTS	Applied Bayesian Modeling, Factor models, Bayesian Nonparametrics, Latent Feature Models, Statistical Ecology.	
CURRENT POSITION	Postdoctoral Associate Duke University , Department of Statistical Science (from 10/2024 to present) Advisors: Amy Herring and David Dunson	
EDUCATION	Università degli Studi di Padova , Padova, Italy Ph.D. in Statistics , Department of Statistical Sciences (from 10/2021 to 09/2024, thesis defense 01/2025) • Thesis Topic: <i>Bayesian methods for complex dependence structures with application in ecology</i> • Advisors: Antonio Canale and David Dunson M.Sc. in Statistical Sciences , Department of Statistical Sciences (from 10/2019 to 09/2021) • Thesis Topic: <i>Bayesian hierarchical models for spatial extreme values</i> • Advisor: Antonio Canale • Final mark: 110/110 with honors B.Sc. in Statistics for Technology and Sciences , Department of Statistical Sciences (from 10/2016 to 07/2019) • Thesis Topic: <i>Quantile regression for solar power forecasting</i> • Advisor: Antonio Canale • Final mark: 110/110 with honors	
WORK EXPERIENCE	• Research support activities (from 03/2021 to 05/2021) Department of Statistical Sciences, Università degli Studi di Padova Supervisor: Prof. Bruno Scarpa • Data scientist intern at Horsa, Vicenza (from 09/2019 to 12/2019) • Ski instructor at Intersport Bernik, Kranjska Gora, Slovenia (from 12/2016 to 2/2020)	
VISITING PERIODS	Duke University , Department of Statistical Science , NC, USA (from 03/2023 to 09/2024) Supervisor: David Dunson .	
AWARDS	Travel & academics awards • First runner up ISBA BioPharm Junior Researcher Award 2025 • Travel award. BNP14 conference, Los Angeles, US (2025) • ASA-SBSS Student Paper Competition (2025) • Best poster award at ISBA 2024 • ISBA travel award. ISBA 2024 world meeting, Venice, Italy • Travel award. 4th Italian Meeting on Probability and Mathematical Statistics (2024) • Best poster award at Autumn school in Bayesian Statistics 2023, CIRM (France) • ISBA travel award. ISBA 2022 world meeting, Montreal, Canada • Mille e una Lode Award 2018/2019, scholarship awarded to the best 3% students of the University of Padova Data competitions • First prize winner at HackTheGene, Padova (09/2022)	

PUBLICATIONS

Submitted and working papers

- Mauri, L. *, **Stolf, F. ***, Herring, A. H. , Miller, C. and Dunson D. B. (2026+). Pathway-based Bayesian factor models for 'omics data. Submitted, *arXiv:2601.13419*.

* denotes equal contribution.

Articles in peer-reviewed journals

- Canale, A., Schiavon, L. and **Stolf, F.** (2026). Identifiable sparse Bayesian factorizations via meta regression. *Statistics & Probability Letters*, 236, 110772.
- **Stolf, F.** and Canale, A. (2026). Bayesian adaptive Tucker decompositions for tensor factorization. *Journal of Computational and Graphical Statistics*, 35(1), 119–128.
- **Stolf, F.** and Dunson, D. B. (2025). Infinite joint species distribution models. *Biometrika*, 112(4), asaf055.
- **Stolf, F.** and Canale, A. (2023). A hierarchical Bayesian non-asymptotic extreme value model for spatial data. *Environmetrics*, 34(7), e2806

Discussions and conference proceedings

- **Stolf, F.** (2025). Dependent infinite latent feature models for tree occurrence data, in *Methodological and Applied Statistics and Demography IV: SIS 2024, Short Papers*.
- Canale, A., Schiavon, L. and **Stolf, F.** (2025). Invited discussion to “Sparse Bayesian factor analysis when the number of factors is unknown”. *Bayesian Analysis*, 20(1), 213–344.
- **Stolf, F.** and Canale, A. (2022). Bayesian spatial modeling of extreme precipitation, in *Proceedings of the 36th International Workshop on Statistical Modelling*. ISBN: 9788855113090

SOFTWARE

- **BASIL**: Bayesian Analysis with gene-Sets Informed Latent space. Development R package. [Github repo](#)

SERVICE TO
PROFESSION

Referee for: Annals of Applied Statistics; Biometrics; Computational Statistics & Data Analysis.

Positions in Academic Societies

Peer mentor in the [j-ISBA Peer Mentoring Scheme](#), “Researchers supporting other researchers” (from 10/2025 to present)

Membership to scientific societies:

- International Society for Bayesian Analysis (ISBA), BNP ISBA and jISBA
- Società Italiana di Statistica (SIS)

Organizer of *Explain like I'm an Undergrad* (from 09/2023 to 07/2024)

Series of weekly seminars for PhD students, post-docs, and young researchers in the statistics department at the University of Padova

PRESENTATIONS

Invited presentations

- [Sep 2025] GRASPA environmental statistics 2025, Rome, Italy
- [Sep 2025] SISBayes 2025, Padova, Italy
- [Aug 2025] JSM 2025: Joint Statistical Meetings, Nashville, USA
- [Dec 2024] CMStats 2024: Computational and Methodological Statistics, London, UK
- [Jun 2024] 4th Italian Meeting on Probability and Mathematical Statistics, Rome, Italy

Contributed presentations

- [Jun 2025] 14th International Conference on Bayesian Nonparametric, Los Angeles, USA
- [Jun 2024] 52th Scientific meeting of the Italian Statistical Society, Bari, Italy
- [Nov 2023] BAYSM 2023: Bayesian Young Statisticians Meeting, online

Poster presentations

- [Jul 2024] ISBA world meeting 2024, Venice, Italy
- [Jun 2024] BAYSM 2024: Bayesian Young Statisticians Meeting, Venice, Italy
- [Oct 2023] Autumn school in Bayesian Statistics 2023, CIRM, Marseille, France
- [Jul 2022] 36th International Workshop on Statistical Modelling (IWSM), Trieste, Italy
- [Jun 2022] ISBA world meeting 2022, Montreal, Canada

TEACHING

Duke University, NC, USA

- Guest lecturer for Bayesian Statistical Modeling and Data Analysis (Ph.D.) - Fall 2025
- Guest lecturer for Bayesian Statistical Modeling and Data Analysis (M.Sc.) - Spring 2025

Università degli Studi di Padova, Padova, Italy

- Teaching assistant for Advanced statistical inference (M.Sc.) - A.Y. 2020/21
- Teaching assistant for Introduction to real analysis (B.Sc.) - A.Y. 2019/20, 2020/21

WORKSHOPS AND SUMMER SCHOOLS

- Selected participant at *Data Research Camp*, Florence (Italy), 07/2024
Four-day meeting where groups of young scholars, advised by a senior researcher, were asked to develop innovative methods and models to analyze a common dataset.
- *Bayes at CIRM: Autumn school in Bayesian Statistics*, Marseilles (France), 10/2023